

INNOVATIVE MINING SYSTEMS

Highwall Mining

By

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Surface Mining

- When ore deposit is large with continuous lateral extension and preferably horizontal or with low dip or if the economics permits this method can be applied even to steeper massive deposits.
- When the ore deposit is at shallow to moderate depth with competent strata and/or orebody whether flatter or steeper so that heavy equipment can be deployed or operated.
- When the ore deposit is preferably uniform i.e. tubular or bedded this method can be applied, if economics permits.
- Applicable in any grade of ore body. Many a times this method is economically viable & technically feasible even if the grade is low.
- When high rate of production and productivity is of paramount importance especially with due regard to conservation, quality and safety.

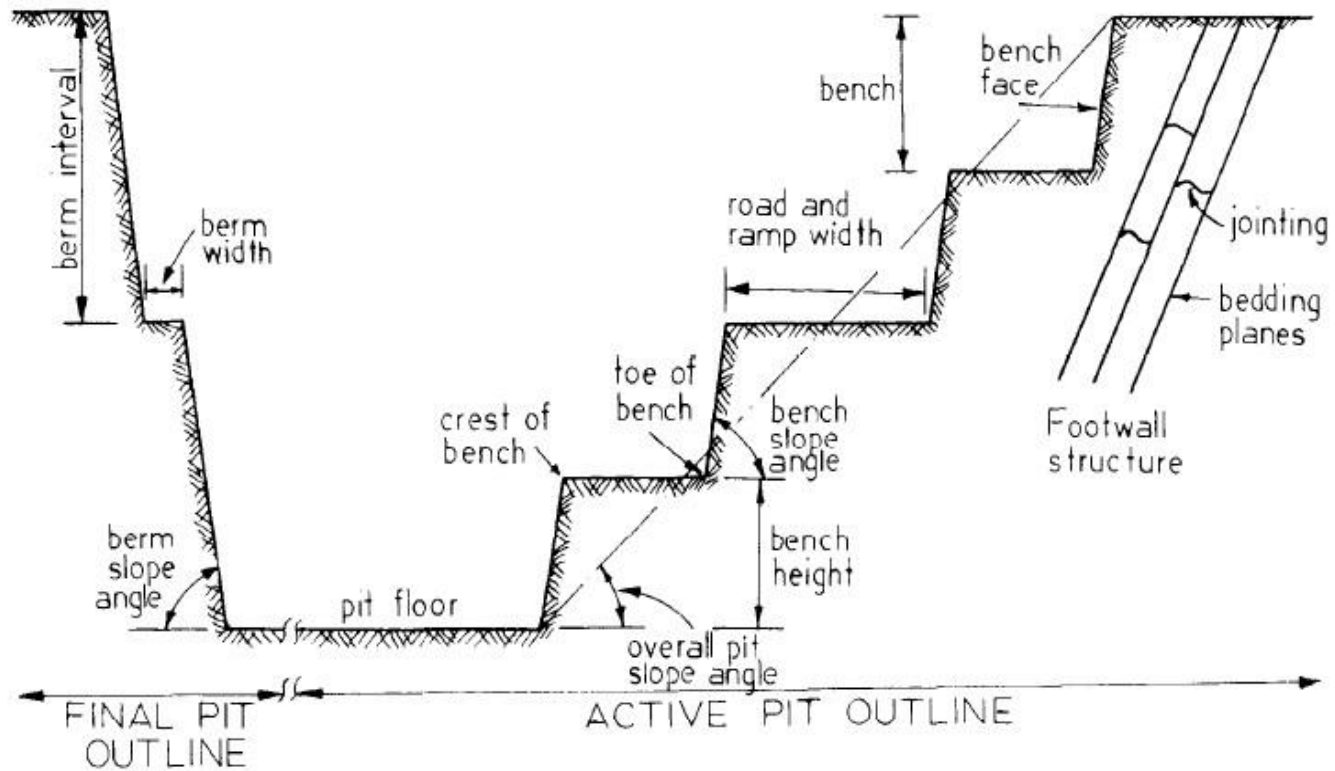
Surface Mining: Advantages

- There is no problem of ventilation (However, very deep surface mines may face ventilation problems.).
- There is no problem of roof control and support.
- There is no problem of lighting in day-time.
- There is no problem of hazardous mine gases.
- Percentage of recovery of coal from seams is much higher (>90%) than that of UG mining (20-60%);
- Higher productivity / OMS (Output per man-shift);
- Mostly profitable venture;
- Danger and mining hazards are lesser in comparison to UG mining;
- Risk of gas explosion or coal dust explosion is negligible;
- Bigger mine size with large scale mechanization and automation is feasible.

Surface Mining: Limitations

- Limited by depth. Beyond such a depth it is to be left for mining in future with upgraded technology of OC mining or may immediately proceed for Underground Mining for extraction in deeper deposits.
- Highly capital intensive.
- Weather detrimental - All season mining is not possible.
- When the ore body or the rock is not competent viz. free flowing argillaceous sand, morture etc. formation of stable benches and the dumps are not possible.
- When Large scale degradation of land in excavation & dump area is not permitted sometimes even with reclamation especially in reserve forest.
- When environmental protection is highly critical OC mining may not be feasible.
- Extremely difficult to safe guard any important and sensitive infrastructure nearby the mine.

Typical open pit Bench Geometry



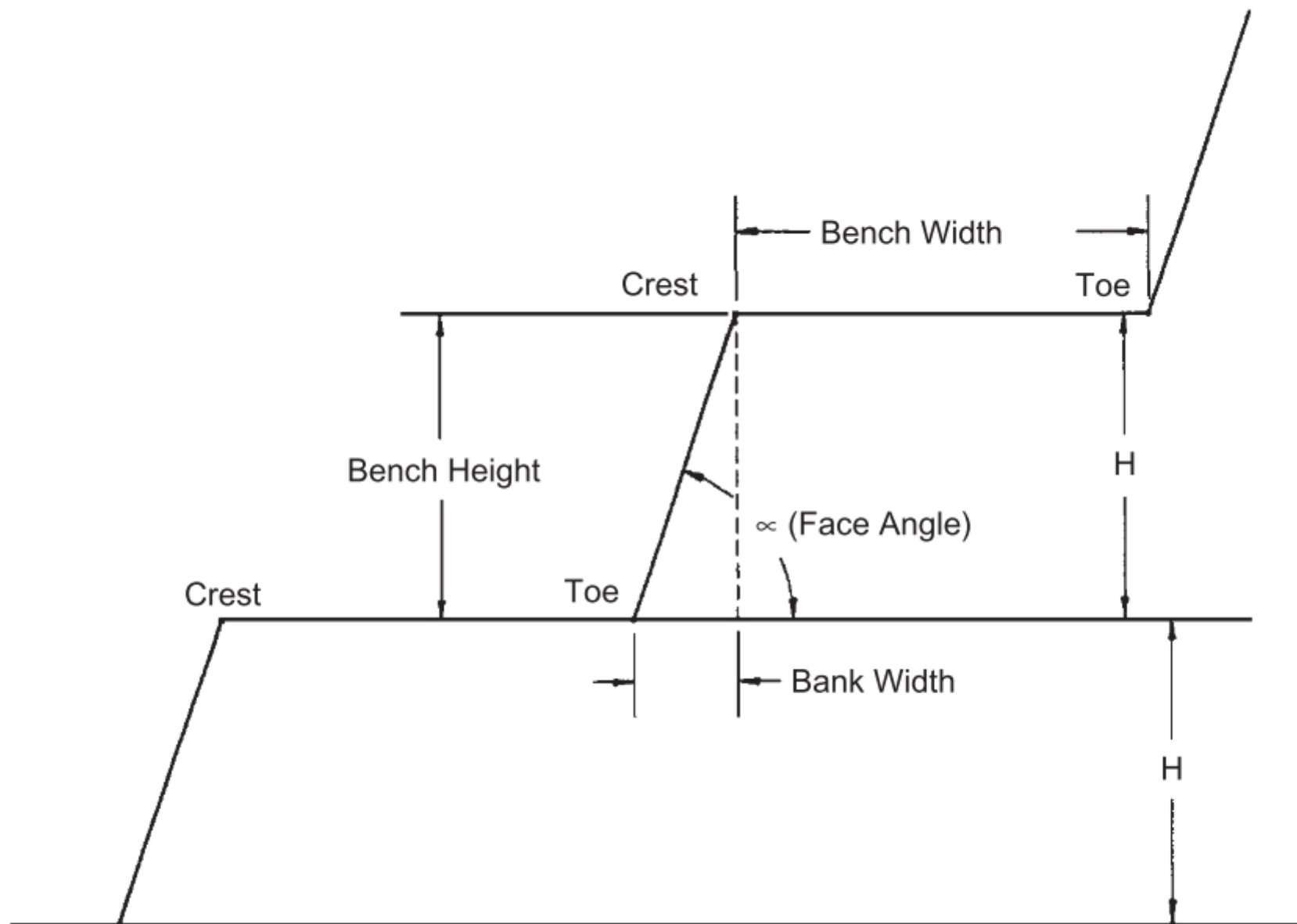


Figure 4.2. Parts of a bench.

17.2.2.1 Computation of overall pit slope angle

Figure¹⁴ 17.3 illustrates the geometry of an open pit. Geometrically overall pit slope angle can be computed as follows:¹⁴

$$OVERALL\ PIT\ SLOPE\ (\phi) = \tan^{-1} \frac{N_B \times B_H}{\{(N_B - 1)B_W\} + \frac{N_B \times B_H}{\tan(B_A)} + (R_W)} \quad (17.4)$$

Whereas: ϕ – overall pit slope angle; degrees

N_B – number of benches

B_H – bench height in meters;

B_W – bench width in meters;

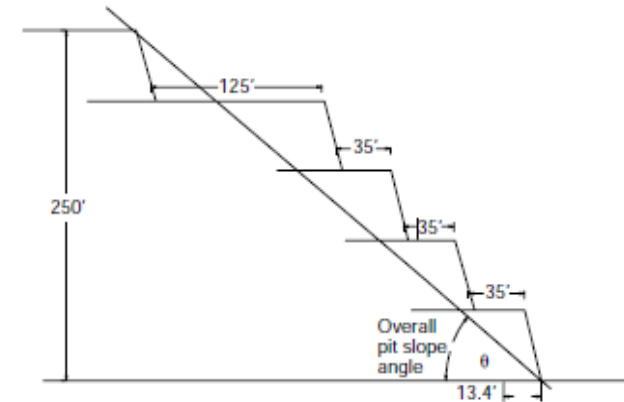
B_A – bench angle in degrees;

R_W – ramp width in meters (if intersected).

Given data: (i) $N_B = 5$; $B_H = 16$ m; $B_W = 12$ m; $B_A = 75^\circ$; $R_W = 0$ m; (answer: 49°)

(ii) $N_B = 8$; $B_H = 10$ m; $B_W = 12$ m; $B_A = 75^\circ$; $R_W = 0$ m; (answer: 37.2°)

(iii) $N_B = 5$; $B_H = 16$ m; $B_W = 12$ m; $B_A = 75^\circ$; $R_W = 30$ m; (answer: 38.8°)



(a) Working pit slope angle is a function of number of benches and bench height

1. A pit slope has the following information :

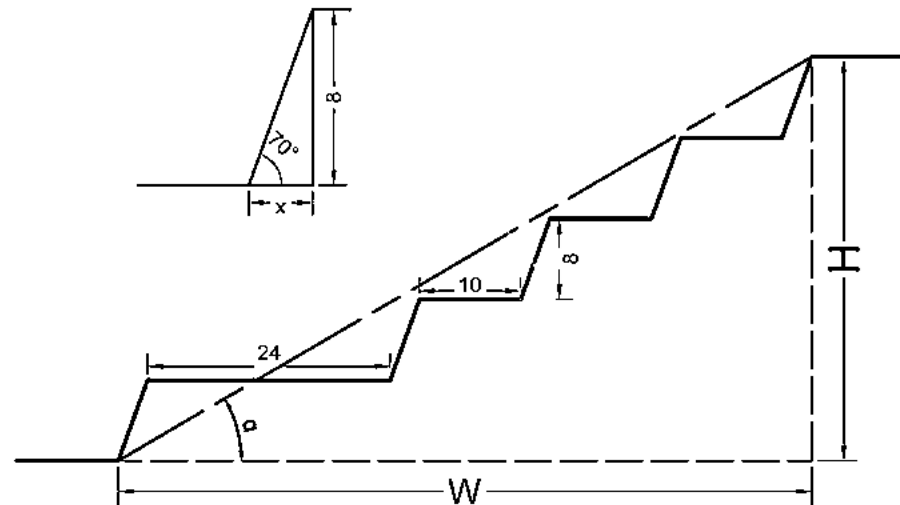
Number of benches = 5

Height of each bench = 8 m

Bench slope angle = 70 degree

If width of one bench is 24 m and that of other four benches is 10 m each, the overall pit slope angle, in degree?

Answer: 30.26 deg



Stripping ratio

- To determine maximum depth based on the profitability of the operation, it is essential to know about the overall costs and revenues that will be received by selling the ore and its by-products, if any.
- In other words what will be cost of removing waste rock that is enclosing the ore body between surface datum and a particular depth, and also mining or exploiting the ore body itself that lies within this envelope.

$$SR_{max} = \frac{(\text{Price or revenues received per ton of ore} - \text{cost of mining per ton. of ore})}{(\text{Waste removing Cost/t})}$$

Highwall Mining

- Highwall mining is a technology that extracts available coal from the highwall faces of the existing opencast mines by deploying remote operated equipment, which have reached their final limit due to uneconomic stripping ratio or due to local constraints (which limit further mining by normal surface/ opencast technology).
- This method relies upon the self supporting capacity of the strata above the series of parallel entries driven mechanically to a considerable depth without artificial roof support and ventilation in the seam horizon.
- This technology provides an economical way to extract coal reserves locked up in the highwall.
- The extent of an opencast project is limited by the financial viability, though coal seam continues to exist beyond the quarry limits.
- Non-replenishable coal is getting lost forever within the highwalls of the opencast projects.
- This technology is being practiced in USA, Australia and Indonesia.
- Limiting gradient – 1 in 3.5 and maximum penetrating depth (in level gradient) is around 500 m.

Highwall Mining

There are basically 2 types of Highwall Miners:

1. Auger Type Highwall Miner (Auger Mining)
2. Continuous type Highwall Miner (Conventional Highwall Mining)

1. Auger Mining:

In Auger Mining, coal is extracted using an auger mining machine. This machine operates like a drill with a cutter head rotating into the coal seam & creating holes to access the coal. The extracted coal is hauled through the auger machine & the conveyor system.

It is comparatively less productive in comparison to Highwall mining system but is safer as its augers (conveying system) are armoured & covered. Auger has also limitation of penetration into the seam.

2. Continuous Mining/ Highwall Mining:

This involves use of a continuous mining machine that creates a rectangular opening in the coal seam of highwall. Coal is hauled to the surface through a conveyor system.

The Technology is highly productive. Production capacity of continuous Mining Machine is up to 1 MT per annum. Continuous Mining Machine can go up to 600m depth.

Highwall Mining

Important characteristics:

1. Highwall mining is a remote controlled mining method of extracting coal reserve that are exposed in the highwall created during surface mining.
2. It extracts coal from the base of an exposed highwall via a series of parallel entries driven to a significant depth within the coal horizon.
3. It allows recovery of coal from surface pits that have reached final highwall position, or in areas where coal has become sterilized, e.g. in-service corridors.
4. It can also be utilized to extract coal left as barrier/ support / un-extractable during conventional underground mining operations.
5. A trench of suitable dimension is dug centrally where Highwall is not available, and parallel drivages are made in both directions along dip as well as along rise.
6. The immediate advantage of highwall mining is that coal reserve can be extracted that would otherwise be uneconomical to mine by conventional surface mining techniques due to high stripping ratio.
7. Since mining of highwall entries is achieved by leaving overburden undisturbed, the systems' economics are independent of strip ratio.
8. During conventional mining it is normally mandatory to acquire land on the surface for carrying out depillaring/final extraction. By this method for ensuring surface stability without any subsidence parting between two openings in the highwall can be suitably designed.

Highwall Mining

Important characteristics:

9. The technology may have wide application in extracting reserves of thin seams (0.8m to 1.5m) which otherwise cannot be extracted economically by conventional technology.
10. The method is capable of making access to such reserves with substantially lesser capital cost and lesser lead time than a conventional underground mine, while being able to produce over 0.6 MT to over one million tonnes of coal per year per system depending on the thickness of seam and depth of Highwall.
11. Highwall/Trench Mining also has a significant operational cost advantage over underground mining, because it is less labour intensive.
12. They remove coal from relatively horizontal seam which are exposed by removing overburden to form a bench or highwall.

Applicability:

Highwall mining technology will have wide application in economically exploiting:

- Reserves confined to thin seams (less than 1.5m) in different coalfields which can not be exploited either by opencast or by conventional underground mining methods economically.
- Reserves blocked in the batter in opencast projects which also can not be extracted because further advancement of quarry is not possible on account of many issues.
- Reserves confined to coal seams lying below the final floor of the quarry at certain depths which cannot be exploited economically.
- Can be deployed to extract coal seams in eco-sensitive area with minimum environmental footprints in cost effective manner.

CIL is planning to deploy 30 HWs by 2029-30. Currently, CIL is operating 5 HWs.

Highwall Mining

Advantages of Highwall mining:

- Safe Mining
- High Production Potential
- Higher Productivity
- No ash dilution
- Conservation of coal
- Easy Mobility
- Lesser capital investment over underground mines

1. Most Safe Method:

- It is the safest method since no man is going inside the drivages.
- Length of the machine is approx. 17m.
- Cabin of the Operator is located at near of the machine & at a height of 8.5m.
- As such no persons are required to work in the potentially danger area i.e. near the Highwall.

Highwall Mining

Advantages of Highwall mining:

2. High Production Potential:

- Production potential of the machines depend upon the thickness of the coal seam and the mining condition of the site.
- Highwall miners can produce approx.:
 - 6200 TPD from thick seams,
 - 4000 TPD from medium seams and
 - 2500 TPD from the thin seams (0.9 to 1.4 m.)
- Production over 100000 Ton per Month per machine is being achieved in a sustained manner from many mines in USA.

3. Higher Productivity:

- The machine is operated with 6-7 crew members per shift with very high productivity.
- Depending upon the thickness of seam, output per man shift can be very high (may be upto 100) which cannot be achieved by any other underground technology.

Highwall Mining

Advantages of Highwall mining:

4. No ash dilution:

- Since the machine is cutting only the coal and being transported inside the closed push beam, coal is not getting contaminated with the fallen roof, if it occurs.
- So there is no ash dilution.

5. Conservation of coal:

- This technology recovers that coal which in normal case, cannot be taken out by any other conventional technology.
- Highwall miners can be deployed to extract coal left in barrier/support pillars/un-extractable area during conventional underground mining operations.
- Depending upon the thickness of coal seam, strength of coal and depth of cover, recovery of coal to the extent of 50% to 70% is possible to achieve.

6. Mobility:

- Machine is mounted on four hydraulically powered crawlers for movement.
- This allows the machine to move both straight and in transverse direction.
- This helps the machine to position itself in the next hole accurately with least time.
- It is modular in construction and can be broken into modules in one day and can be assembled in 3-4 days.

7. Lesser capital investment over underground mines:

- Capital investment in this method is much less compared to mining.

Highwall Mining block selection

- The site investigations for identification of suitability of highwall mining blocks must consider the geological characterisation.
- This must be followed by the review of mine plan of the site and long-term planning.
- In some cases, if mine management wants to extend the open pit in any other directions other than the highwall mining side, then consideration must be given to long-term stability of the highwall mining excavations, and the logistics aspects.
- In the case of multiple seam highwall mining, backfilling of the pit to make a platform for driving excavations into the upper seams is necessary.
- The cost associated with planning and logistics must be estimated properly considering the constraints like power consumption and time factor, especially when multiple seam highwall mining is considered for operation.
- Sufficient contingencies must be put in place, such as overhead for delays in implementation in logistics and planning.
- Provision of enough budget shall be put aside for basic site investigation, second-pass investigation for detail characterization and design, as well as for additional investigations. Detailed economic viability should be carried out, incorporating costs associated with site investigations, monitoring, site preparation, pit backfilling for multiple seam highwall mining, power, dewatering, site logistics, equipment, etc.

Highwall Mining block selection

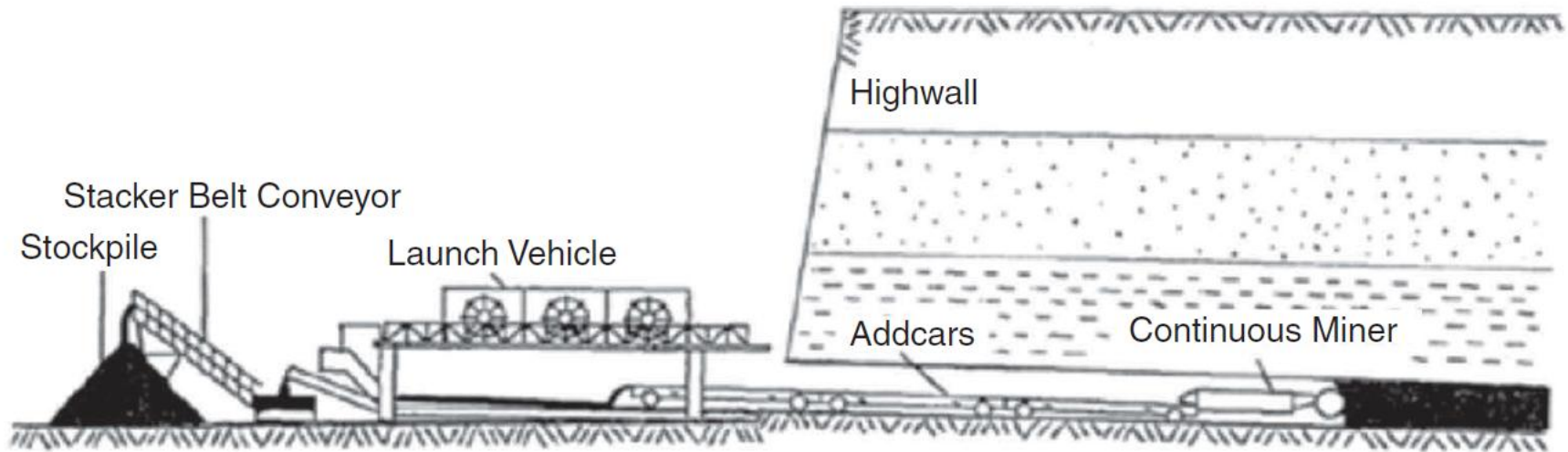
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Highwall Mining block selection

Among many geological features, the following are quite important for the highwall mining design and operations:

- Seam conditions – dip, rolls, thickness variations, strength, methane/CO₂ emissions,
- cleat partings, sandstone or hard band lenses within the coal seams, structure, groundwater, etc.
- Roof and floor conditions – strengths, presence of clay bands and shales, bedding
- planes and their thicknesses, roof/seam interface, structures, lithology and stratigraphy, etc.
- Highwall – slope angle, structures and formation of planar, wedge or toppling or combination of these, blast impacts and damages due to weathering, groundwater run offs, etc.
- Overburden and surface conditions – overburden thicknesses and its densities, topography, dumps, rivers, canals, villages, agriculture lands and forestlands, railway lines, monuments, etc.
- Geological structures – faults, folds, dyke intrusions, etc..

Highwall Mining



Highwall Mining



Highwall Mining



Web cut



Highwall miner



Highwall miner



Trench Mining



Auger Mining



High wall Mining Technology – Auger Type

- Consisting of single or dual cutting heads with coal being cleared by spiraled flights, creating circular entries



Question:

Coal seam is being extracted by highwall mining. Between two successive web cuts, a rectangular web pillar is left to maintain the stability of the overlying strata.

The following data are given:

Shorter width of web pillar = 5 m

Longer width of web pillar = 100 m

Working height = 4 m

Depth of cover = 60 m

Uniaxial compressive strength of coal = 14 MPa

Web cut width = 3.5 m

Unit rock pressure = 0.025 MPa/m

Assume that:

The tributary area method is applicable for load estimation.

The web pillar behaves as a long pillar.

Sheorey's empirical pillar strength equation is valid.

(a) Determine the equivalent width w_e of the web pillar.

(b) Check whether the condition $\frac{w}{h} > 1.0$ is satisfied and comment on whether any strength reduction is required.

(c) Calculate the pillar strength S using Sheorey's empirical equation

(d) Estimate the load on the web pillar P using the tributary area method:

(e) Determine the factor of safety (FOS) of the web pillar.

(f) Comment on the stability of the pillar (short-term / medium-term / long-term) based on Indian coal mining practice.

Design formulations for web pillar and entry span

- For pillar design (web or barrier pillars), one needs to judge the pillar based on laboratory and field experiments, and numerical modelling, as well as empirical and experiential approaches.
- In-depth analysis of case studies about their failure has revealed that accurate web pillar design is the key to successful highwall mining operations.
- Web pillar design involves three major steps viz., estimation of web pillar strength, estimation of load on web pillar and determination of Factors of Safety (FoS).
- A study of underground stone mines found that the strength of slender pillars that is W/h ratio less than two is highly variable and follows a different failure process compared to traditional square pillars (Esterhuizen, 2006).
- Thus, countries practicing highwall mining have developed their own pillar strength formulae suitable for highwall mining scenarios.

3.2.2 Salamon and Munro formula

Salamon and Munro (1967) carried out an intensive investigation into the strength of rectangular shaped coal pillars by statistically analysing 96 intact and 27 collapsed pillar geometries in South African coal mines.

Australian mining conditions are well represented by Salamon-Munro equation. The generalised equation is given below:

$$S = K_0 . h^\alpha . w^\beta \quad (3.2)$$

where,

S – pillar strength, MPa

K_0 – mass cube strength = 7.176 kPa

h – pillar height (m), w is the width of a square or rectangular pillar (m)

$\alpha = -0.66$ and $\beta = 0.46$.

The main reason for the popularity of the Salamon and Munro (1967) equation is that the data used was from the mines and the strength was taken as the mean strength of coal pillars, as opposed to the strength of coal specimens. The strength equation is applicable to slender pillars. For evaluation of the long-term stability of Indian coal pillars, Sheorey et al. (1986) generalised the Salamon and Munro (1967) equation as:

$$S = 0.79K . w^{0.46} / h^{0.66} \quad (3.3)$$

where K (cube strength) needs to be evaluated by the testing of specimens 30 cm in size. If no data exists regarding compressive strength of large specimens, the traditional approach is to apply a reduction factor to the USC estimated from small rock samples (Vandergrift and Garcia, 2005), in an order that ranges from 30–70% (Iannacchione, 1999).

Over the period of last three decades, there have been many pillar strength formulae proposed by various authors. Among them a few formulae that have been used in the design of web or barrier pillars are summarised below.

3.2.4 CSIR-CIMFR pillar strength formula

The CSIR-CIMFR pillar strength equation has been developed over a couple of decades after analysing a large number of pillar stability observations from a gamut of Indian mining scenarios. The CSIR-CIMFR pillar strength equation developed by Sheorey (1992) is given below:

$$S = 0.27 * \sigma_c * h^{-0.36} + \left(\frac{H}{250} + 1 \right) \left(\frac{W_e}{h} - 1 \right) \quad (3.5)$$

where,

S is web pillar or barrier pillar strength (MPa)

H is the cover depth (m)

W_e is the equivalent pillar width (m) = $2W$ for long pillar

W is the width of the web pillar (m)

h is the working mining entry height (m)

σ_c is the strength of 25 mm cube coal sample (MPa)

Any deviation in pillar width will lead to a reduction of pillar strength and thereby its safety factor (CSIR-CIMFR, 2008). Thus, to account for deviation in pillar width, the following correction is applied and accordingly pillar strength is computed.

$$W_d = W \pm 2dL \quad (3.6)$$

where,

W_d = deviated width (m)

W = web pillar width (m)

d = percentage deviation with respect to length of web cut

L = maximum length of extraction cut (m)

Effective width

$$w_e = \frac{4A}{O} = \frac{2w_i w_s}{(w_i + w_s)} = \frac{2w_s}{\left(1 + \frac{w_s}{w_i}\right)} = 2w_s \text{ as } w_s \ll w_i$$

A = Plane area of the web pillar in m^2 , O = Perimeter of the web pillar in m, w_l = Longer width of the web pillar in m, w_s = Shorter width of the web pillar in m.

Calculate effective width using the formula for 8 m width highwall web pillar and 200 m long

Ans: 15.38 m

When the **w/h ratio is less than or close to 1.0**, to err on the safer side, the strength using equation (1) using **$w_e = 2w$** needs to be reduced by **20%**.

Factor of safety (FOS)

$$\text{FOS} = \frac{\text{Strength of pillar}}{\text{Load on pillar}} = \frac{S}{P}$$

Load on web pillars is estimated using the tributary area method, which is given as:

Load on pillars is generally estimated using the tributary area theory, from the following equation:

$$P = \frac{\gamma H (W + W_c)}{W} \quad (3.12)$$

where P is the load on web pillar (MPa); γ is the unit rock pressure (MPa/m); H is the depth of cover; W is the width of web pillar and W_c is the width of web cut. However the load ' P ' is usually not uniform throughout the length of web pillars due to various other factors, such as depth variation, end effect, or induced abutment stresses caused by pit or trench excavation, thus it is prudent to obtain the maximum vertical stress acting on a web pillar using numerical modelling.

Based on past experiences from Indian coalfields, it has been observed that a **pillar safety factor of more than or equal to 2.0** is **long-term stable**, i.e., stable for many decades.

A **FOS between 1.5 to 2** may be taken as **medium-term stable**, stable for a few years. If the **FOS of the pillar is 1.0-1.5**, it may be treated as **short-term stable**, with a stand-up time of a few weeks or a month.

FOS < 1	unstable
FOS 1-1.5	short term stable
FOS 1.5-2	medium term stable
FOS >2	long term stable

Equivalent width: 9.52

Pillar strength: 4 Mpa

Load on pillar: 2.55

FOS: 1.54

Stability: medium term stable

For 6 m wide web pillar FOS: 1.9

Stability: medium term stable

For 7 m wide web pillar FOS: 2.1

Stability: medium term stable

To calculate **percentage of extraction** for web pillar mining, we use the **plan-area basis**:

$$\text{Percentage of extraction} = \frac{\text{Width of cut}}{\text{Width of cut} + \text{Width of web pillar}} \times 100$$

Percentage of extraction				
	Strike length	Width of cut	Width of web pillar	Percentage of extraction
	800	3.5	8	30.43
	800	3.5	9	28.00
	800	3.5	10	25.93
	800	3.5	11	24.14
	800	3.5	1.5	70.00

Using the data provided in Table 1, calculate the total ore tonnage, assuming an ore density of 2.27 t/m³ and assume trapezoidal rule. Note that the extracted ore volume includes 110,495 m³ from old workings and 424 m³ extracted during the previous year. Considering a 70% recovery, compute the recoverable ore tonnage. Further, assuming that the total overburden (OB) volume to be excavated is 2,442,188 m³, calculate the all-time stripping ratio (m cube per tonne).

Table 1

Section line	Sectional area	Length of
	(m ²)	influence (m)
Boundary1	0	0
AA'	11275	57
BB'	19266	51
CC'	9485	57
DD'	3971	54
EE'	2517	61
Boundary2	350	55

Answer:

End Area Formula (Trapezoidal Rule)

This is also known as trapezoidal formula. The volume is assumed to be equal to the average of the areas of the two ends multiplied by the distance between them.

Volume between first two sections, $V_1 = \frac{(A_1 + A_2)}{2} L$

Volume between next two sections, $V_2 = \frac{(A_2 + A_3)}{2} L$ and so on

Total Volume, $V = V_1 + V_2 + \dots + V_n$

$$V = L \left[\frac{A_1 + A_n}{2} + A_2 + A_3 + \dots + A_{n-1} \right]$$

where, $A_1, A_2, A_3, \dots, A_n$ are the end consecutive areas and L is the distance between them.

Interval	(A1) (m ²)	(A2) (m ²)	Length (m)	Volume (m ³)
Boundary1 – AA'	0	11,275	57	321,338
AA' – BB'	11,275	19,266	51	778,051
BB' – CC'	19,266	9,485	57	819,198
CC' – DD'	9,485	3,971	54	363,312
DD' – EE'	3,971	2,517	61	198,884
EE' – Boundary2	2,517	350	55	78,843

A-Gross ore volume (Table 1): 2559626 m³

B-Extracted (od + last year): 110919 m³

Remaining volume (A-B): 2448707 m³

Total tonnage: 2448707 m³ * 2.27 = 5558566 tonne

70%recovery

Recoverable ore=0.70×5,558,566 = 3891000

SR=2,442,1883,891,000

SR = 0.63